

Structured Knowledge Accumulation: Geodesic Learning Paths and Architecture Discovery in Riemannian Neural Fields

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Abstract

The Structured Knowledge Accumulation (SKA) framework redefines neural learning as a process of entropy-guided knowledge organization. This paper extends SKA to continuous neural fields using Riemannian geometry, introducing *Riemannian SKA Neural Fields*. By modeling the neural medium as an information manifold with a metric tensor encoding entropy and density gradients, we enable knowledge propagation along geodesics, balancing information structuring with spatial heterogeneity. This approach generalizes the entropic least action principle to spatially continuous systems, unifying architecture discovery with learning through entropy minimization, and offering a biologically plausible and computationally efficient paradigm for adaptive learning. We employ finite element methods and discrete exterior calculus to implement the framework, establishing theoretical foundations for scalability to higher-dimensional spaces. This work bridges information theory, differential geometry, and neural computation, opening new avenues for real-time, resource-efficient AI systems.

1 Introduction

The Structured Knowledge Accumulation (SKA) framework, as developed in our previous work [1, 2], established a fundamentally new approach to learning in neural systems. In this paradigm, learning is governed by the continuous-time local reduction of entropy through the progressive alignment of knowledge and decision probability shifts, achieved via forward-only dynamics. The SKA formulation provided a rigorous, biologically plausible, and computationally efficient principle for knowledge accumulation in both artificial and natural neural architectures.

However, the original SKA framework and its subsequent temporal generalization were developed primarily for layered systems in which learning unfolds as a sequence of forward steps. While this perspective offers clear insights into local knowledge structuring and entropy dynamics, it does not capture the complexities introduced by spatially continuous systems with varying neuron density, anisotropic connectivity, or task-dependent functional organization. Here, we aim to extend the SKA paradigm by exploring the transition from discrete layered networks to a continuous neural field formulation, thereby enabling a unified description of both spatially structured and functionally organized learning.

In this work, we introduce the *Riemannian SKA Neural Fields* framework—a principled geometric generalization of SKA that addresses this foundational gap. Here, the neural field is formalized as an information manifold, endowed with a Riemannian metric tensor that intrinsically encodes both entropy gradients and structural (e.g., density) gradients. Knowledge accumulation is reinterpreted as propagation along geodesics of this manifold, naturally balancing information structuring with system architecture. This approach transcends the limitations of previous SKA formulations, eliminating the need for ad hoc probabilistic connectivity and

providing a mathematically unified foundation for learning in complex, spatially or functionally heterogeneous neural systems.

By bridging information theory, differential geometry, and biological realism, Riemannian SKA Neural Fields establish a new paradigm for the analysis and design of adaptive learning systems. This work generalizes the principle of entropic least action from time to space and function, offering deep theoretical insight and novel opportunities for scalable, efficient neural computation.

2 Literature Review

The Riemannian SKA Neural Fields framework builds upon and extends several foundational areas of research. This section positions our contribution within the existing literature on neural field models, information geometry, Riemannian approaches to neural networks, architecture discovery, and discrete computational geometry.

2.1 Neural Field Models

Continuous neural field theories provide the foundation for modeling spatially extended neural systems:

- **Amari (1977)** [5]: Introduced continuous neural dynamics in lateral-inhibition type neural fields, establishing pattern formation through differential equations governing neural activity. This seminal work demonstrated how continuous spatial models can capture emergent neural phenomena.
- **Kohonen (1990)** [10]: Developed self-organizing maps that learn topological representations through competitive learning in continuous neural sheets, demonstrating how spatial organization emerges from local interactions.
- **Gap in existing work:** While these models describe neural activity dynamics in continuous space, they require separate, externally defined learning rules. None formulate learning itself as an intrinsic continuous process driven by entropy minimization through forward-only dynamics.

2.2 Information Geometry in Learning

Information geometry provides powerful mathematical tools for understanding learning systems:

- **Amari & Nagaoka (2000)** [6]: Established the mathematical framework of information geometry, demonstrating how the Fisher information metric defines natural gradients in parameter space. This work showed that gradient descent can be improved by accounting for the geometric structure of probability distributions.
- **Nielsen (2020)** [18]: Provided an accessible introduction to information geometry, emphasizing its applications to machine learning and statistical inference.
- **Key distinction:** While Amari’s natural gradient uses the Fisher information metric on *parameter space*, our framework employs entropy and density gradients to define a metric on *physical neural space*. The Fisher metric describes statistical geometry of parametric models, whereas our entropy-based metric emerges directly from the information-theoretic structure of knowledge accumulation in spatially distributed neural media.

2.3 Riemannian Geometry in Neural Networks

Recent work has explored geometric perspectives on neural learning:

- **Bronstein et al. (2017)** [8]: Developed geometric deep learning, extending neural networks to non-Euclidean domains such as graphs and manifolds. This work demonstrated the power of geometric inductive biases.
- **Benfenati & Marta (2022)** [12]: Applied Riemannian geometry to deep neural network analysis, providing theoretical foundations for understanding learning dynamics in curved spaces.
- **Katsman et al. (2023)** [34]: Introduced Riemannian residual neural networks, demonstrating performance improvements by incorporating geometric structure into network architecture.
- **Simon et al. (2021)** [35]: Explored geodesic paths for incremental learning, showing how geometric interpolation can improve continual learning performance.
- **Chen et al. (2018)** [38]: Introduced Neural ODEs, demonstrating how continuous-depth models can be formulated as differential equations, opening new directions for understanding neural dynamics.
- **Our contribution:** Existing approaches *impose* Riemannian structure to improve optimization or generalization. In contrast, our metric tensor *emerges naturally* from the entropy and density landscapes of the learning system itself. The geometry is not a design choice but an intrinsic property of entropy-driven knowledge accumulation. Furthermore, we eliminate backpropagation entirely through forward-only dynamics along geodesics.

2.4 Architecture Discovery

Automated neural architecture search has become a major research direction:

- **Neural Architecture Search (NAS)** [14, 15, 16, 17]: These methods search discrete architecture spaces using reinforcement learning, evolutionary algorithms, or differentiable relaxations. NAS has achieved state-of-the-art results but requires extensive computational resources and operates on predefined search spaces.
- **Our approach:** Rather than searching discrete architecture spaces, we discover optimal connectivity patterns *continuously* through geodesic computation on the information manifold. Architecture emerges from entropy minimization rather than external search. The neural field self-organizes connectivity by following information-optimal paths, requiring no predefined search space or discrete optimization.

2.5 Discrete Exterior Calculus in Neural Systems

Discrete exterior calculus provides computational tools for geometric learning:

- **PyDEC (Bell & Hirani, 2011)** [13]: Developed software infrastructure for discrete exterior calculus, enabling computational differential geometry on simplicial complexes.
- **Desbrun et al. (2005)** [33]: Established theoretical foundations of discrete exterior calculus, showing how continuous differential forms can be faithfully discretized while preserving geometric structure.

- **Applications** [32, 36]: Recent work has applied DEC to physics-informed machine learning and scientific computing, demonstrating its power for enforcing geometric constraints.
- **Zhang et al. (2023)** [31]: Introduced neural representations for fast geodesic distance and path queries, demonstrating computational efficiency of geometric methods in neural systems.
- **Our use:** We are the first to apply discrete exterior calculus to *neural learning path computation*, using DEC to compute geodesics that represent optimal knowledge propagation routes in continuous neural fields. This enables efficient, geometrically principled implementation of the theoretical framework.

2.6 Biological Neural Organization

Understanding biological neural systems informs our geometric approach:

- **Collins et al. (2010)** [3]: Documented neuron density variations across cortical areas in primates, providing empirical foundation for our variable-density neural field model.
- **Hebb (1949), Bi & Poo (1998)** [19, 20]: Established activity-dependent plasticity principles. Our framework shows these emerge naturally from entropy minimization rather than requiring separate learning rules.
- **Brain-inspired architectures** [21, 22, 23, 28]: Recent neuromorphic systems demonstrate the importance of spatial structure and connectivity patterns. Our geodesic-based connectivity provides theoretical foundation for these empirical observations.

Furthermore, Riemannian SKA Neural Fields derive key principles of biological neural economy from first principles. The emergent geodesic connectivity recovers the empirically observed wire-optimization principle [39] as a direct consequence of entropic least action, providing a physical mechanism for why biological connectomes are near-optimally economical. Similarly, it derives the empirically observed economical small-world organization of biological connectomes [40] as the steady-state solution of entropic geodesic flow, unifying functional and metabolic optimization under a single information-theoretic principle.

2.7 Positioning the Contribution

The Riemannian SKA Neural Fields framework synthesizes these research directions while making several novel contributions:

1. **Intrinsic geometry:** Unlike imposed geometric structures, our metric emerges directly from entropy and neuron density gradients.
2. **Geodesic knowledge propagation:** Knowledge propagates along spatial geodesics, extending the entropic least action principle of SKA [1, 2] to continuous spatial fields. Each location updates based on local entropy gradients and incoming information from neighbors, enabling efficient computation on curved information manifolds.
3. **Continuous architecture discovery:** Rather than discrete search, connectivity patterns emerge continuously via geodesic computation.
4. **Unified temporal and spatial principles:** Local learning follows temporal entropic least action (Papers I-II), while global propagation follows spatial least action on the information manifold.

5. **Dimensional scalability:** The framework naturally extends to arbitrary spatial dimensions (4D, 5D, etc.) without modification to core equations.
6. **Computational implementation:** Integration of FEM [7], DEC [13, 33], and Riemannian geometry [4] provides practical tools for realizing the theoretical framework.

This work thus establishes a principled geometric foundation for continuous neural learning that unifies information theory, differential geometry, and biological realism.

3 From Discrete to Continuous Neural Representations

3.1 Review of the SKA Framework

In the SKA framework [1, 2], learning occurs through the alignment of knowledge tensors with decision probability shifts, governed by the principle of entropic least action. The layer-wise entropy is defined as:

$$H^{(l)} = -\frac{1}{\ln 2} \sum_{k=1}^K \mathbf{z}_k^{(l)} \cdot \Delta \mathbf{D}_k^{(l)} \quad (1)$$

This formulation captures how structured knowledge accumulates as neural networks learn to reduce uncertainty through forward-only information flow.

3.2 Neuron Density and Continuous Fields

To extend SKA to a continuous field theory, we introduce the concept of neuron density $\rho(\mathbf{r})$, representing the number of neurons per unit volume at position $\mathbf{r} = (x, y, z)$. In the primate brain, this density varies significantly across regions, from approximately 30,000 to 180,000 neurons per cubic millimeter [3].

For each infinitesimal volume element dv at position $\mathbf{r}$, the number of neurons is given by:

$$n(\mathbf{r}) = \rho(\mathbf{r}) \cdot dv \quad (2)$$

This neuron count determines the dimensionality of the knowledge and decision probability tensors at that point in space. To simulate the spatial distribution of neurons in our continuous neural field model, we implement a computational approach using Simplex noise from the `noise` library for both 3D and higher-dimensional models to create natural, smoothly varying density patterns. This noise function produces biologically-inspired neuron distributions with localized variations while maintaining overall continuity, as shown in Figure 1. The statistical properties of Simplex noise allow us to generate realistic density fields that resemble the natural variability observed in biological neural tissues across different dimensionalities.

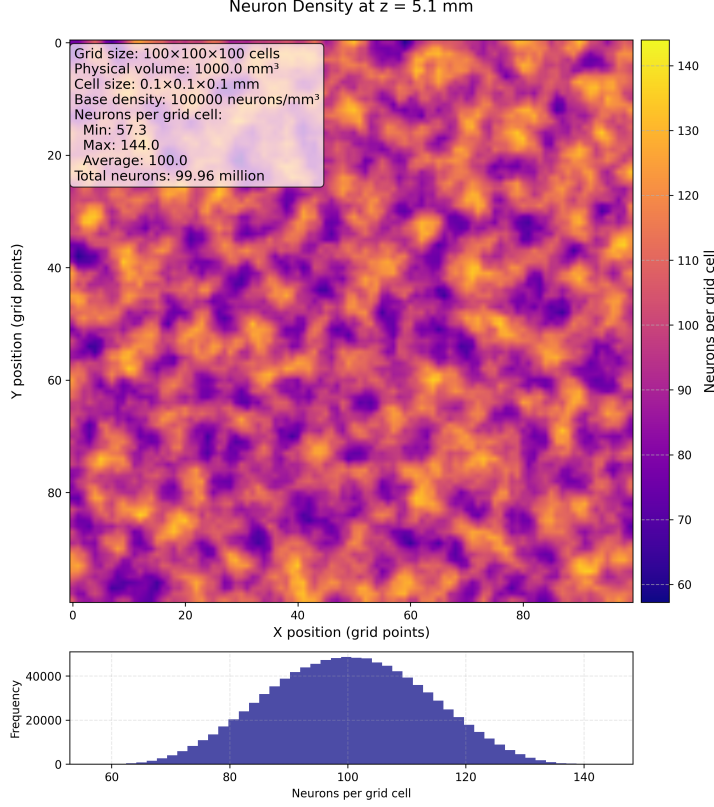


Figure 1: Heatmap of neuron density per grid cell in a $100 \times 100 \times 100$ cubic grid (cell volume = 0.001 mm^3) with a baseline density of $100,000 \text{ neurons/mm}^3$ modulated by Perlin noise ($\pm 50\%$ variability). A colorbar shows the neuron count per cell, and a histogram below illustrates the distribution across the grid.

3.3 Knowledge and Decision Fields as Tensors

We define the following tensor fields for our continuous neural medium:

- $\mathbf{X}(\mathbf{r}, t)$: Input tensor field of dimension d
- $\mathbf{W}(\mathbf{r}, t)$: Weight tensor field with dimensions $n(\mathbf{r}) \times d$
- $\mathbf{b}(\mathbf{r}, t)$: Bias tensor field of dimension $n(\mathbf{r})$
- $\mathbf{Z}(\mathbf{r}, t)$: Knowledge tensor field of dimension $n(\mathbf{r})$
- $\mathbf{D}(\mathbf{r}, t)$: Decision probability tensor field of dimension $n(\mathbf{r})$
- $\Phi(\mathbf{r}, t)$: Knowledge flow tensor field of dimension $n(\mathbf{r})$

Here, $\mathbf{r}$ represents a position vector in a D -dimensional space (with $D=3$ for our initial implementation, but generalizable to higher dimensions $D=4$, $D=5$, etc.). These quantities exhibit dual dimensionality: they exist in D -dimensional physical space while at each spatial point, they contain representational dimensions that vary according to local neuron density $n(\mathbf{r})$. This creates a mathematically rich structure where tensors must account for both spatial coordinates and representational dimensions simultaneously.

The entropy gradient tensor $\nabla \mathbf{h}(\mathbf{r}, t)$ presents a special case. While knowledge and decision tensors have representational dimensions that vary with neuron density, the entropy gradient maintains a consistent dimensionality equal to the spatial dimensionality D , calculated from

scalar entropy differences between neighboring points. This dimensional invariance makes it the natural guide for directing connectivity throughout the neural field, regardless of the ambient space dimension.

The tensor representation allows us to precisely track how these dual-natured tensors interact and evolve throughout spaces of arbitrary dimension. As knowledge tensors propagate through regions of different neuron densities, their interactions are governed by the principle of entropic least action and the geodesic connectivity framework that enables information to flow along entropy-minimizing paths.

4 Local Computation and Knowledge Propagation

4.1 Local Neural Computation

For each volume element at position $\mathbf{r}$, the initial knowledge state is computed as:

$$\mathbf{Z}_0(\mathbf{r}) = \mathbf{W}(\mathbf{r}) \cdot \mathbf{X}(\mathbf{r}) + \mathbf{b}(\mathbf{r}) \quad (3)$$

This creates a direct parallel with the discrete SKA framework, where each volume element functions as a complete neural computation unit that:

- Takes in external inputs $\mathbf{X}$
- Processes them through local weights $\mathbf{W}$ and biases $\mathbf{b}$
- Generates an initial knowledge state $\mathbf{Z}_0$
- Evolves this knowledge according to the entropic least action principle

4.2 Spatial Connectivity

The spatial structure of the neural field emerges from local interactions between adjacent volume elements. The output from one region becomes the input to neighboring regions through:

$$\mathbf{X}(\mathbf{r} + d\mathbf{r}) = \mathbf{D}(\mathbf{r}) = \sigma(\mathbf{Z}(\mathbf{r})) \quad (4)$$

This creates a forward-only flow of information through the neural medium, preserving the unidirectional property of the SKA framework.

4.3 Temporal Evolution

The knowledge field evolves over time according to:

$$\mathbf{Z}(\mathbf{r}, T) = \mathbf{Z}(\mathbf{r}, 0) + \int_0^T \Phi(\mathbf{r}, t) dt \quad (5)$$

Where $\Phi(\mathbf{r}, t)$ is the knowledge flow field at point $\mathbf{r}$ and time t . The decision probability field is derived from the knowledge field through:

$$\mathbf{D}(\mathbf{r}, T) = \sigma(\mathbf{Z}(\mathbf{r}, T)) \quad (6)$$

This time-dependent formulation allows us to:

- Track how knowledge evolves over time in each volume element
- Model the propagation of knowledge waves through the neural medium
- Identify characteristic timescales for knowledge accumulation in different regions
- Study the emergence of structured knowledge patterns over time

5 Entropy Density and the Principle of Entropic Least Action

5.1 Local Entropy Density

For each volume element, we define the local entropy density as:

$$h(\mathbf{r}) = -\frac{1}{\ln 2} \mathbf{z}(\mathbf{r}) \cdot d\mathbf{D}(\mathbf{r}) \quad (7)$$

This represents the entropy contribution from the volume element at position $\mathbf{r}$. The dimension of tensors $\mathbf{z}$ and $d\mathbf{D}$ corresponds to the number of neurons in the infinitesimal volume element at that point.

Note that since $\mathbf{z}(\mathbf{r})$ has length proportional to the local neuron density $\rho(\mathbf{r})$, this definition implicitly incorporates neuron density—regions of higher ρ yield proportionally larger $\mathbf{z}$, so for a fixed change in $d\mathbf{D}$ the entropy per volume naturally decreases as density increases.

5.2 Field Equations from Entropic Least Action

The principle of entropic least action governs how the knowledge field evolves over time. The weight update equations in the continuous field are:

$$\frac{\partial \mathbf{W}(\mathbf{r}, t)}{\partial t} = \frac{1}{\ln 2} \mathbf{z}(\mathbf{r}, t) \odot \mathbf{D}'(\mathbf{r}, t) \otimes \mathbf{X}(\mathbf{r}, t) \quad (8)$$

For regions receiving input from neighbors:

$$\frac{\partial \mathbf{W}(\mathbf{r}, t)}{\partial t} = \frac{1}{\ln 2} \mathbf{z}(\mathbf{r}, t) \odot \mathbf{D}'(\mathbf{r}, t) \otimes \mathbf{D}(\mathbf{r} - d\mathbf{r}, t) \quad (9)$$

Where:

- $\mathbf{D}'(\mathbf{r}, t) = \mathbf{D}(\mathbf{r}, t) \odot (1 - \mathbf{D}(\mathbf{r}, t))$ is the derivative of the decision field with respect to the knowledge $\mathbf{z}(\mathbf{r}, t)$
- $\odot$ represents element-wise multiplication
- $\otimes$ represents the outer product operation

5.3 Universal Learning Equations

A remarkable property of the entropic least action principle is that these field equations are identical to those governing discrete SKA neural networks [1, 2]. In the discrete case, the spatial coordinate $\mathbf{r}$ simply represents layer indices rather than continuous positions, but the mathematical form remains unchanged. This universality arises because neuron density modifies the equations transparently by changing tensor dimensions while preserving the invariant structure of mathematical operations.

This universality suggests that entropic least action produces architecture-independent learning laws that transcend the discrete-continuous boundary. The same fundamental equations govern information processing whether implemented in layered networks or continuous neural fields, revealing deep mathematical unity in systems that minimize entropy through structured knowledge accumulation. This demonstrates that discrete neural networks are approximations of continuous information manifolds and provides theoretical justification for why discrete networks work—they're discretizations of fundamental continuous processes.

6 Numerical Foundations of 3D SKA Neural Fields

To implement the continuous neural field theory in a computational framework, we discretize the spatial domain while preserving the theoretical principles of SKA. Each position in the neural medium is represented through a grid-based system that maintains local computational properties while enabling efficient numerical simulation.

6.1 Spatial Discretization

We define a three-dimensional computational grid where position indices (i, j, k) map to physical coordinates (x, y, z) through a scaling factor s :

$$x = i \cdot s \quad (10)$$

$$y = j \cdot s \quad (11)$$

$$z = k \cdot s \quad (12)$$

Each grid point (i, j, k) represents a local volume element with physical dimensions s^3 containing $n(i, j, k) = \rho(i, j, k) \cdot s^3$ neurons, where $\rho(i, j, k)$ is the neuron density at that position.

6.2 Local SKA Representation

At each grid position (i, j, k) , we define a local SKA computation unit characterized by:

- Weight tensor $\mathbf{W}(i, j, k)$ with dimensions $n(i, j, k) \times d$, where d is the input dimension
- Bias tensor $\mathbf{b}(i, j, k)$ with dimension $n(i, j, k)$
- Knowledge tensor $\mathbf{Z}(i, j, k)$ with dimension $n(i, j, k)$
- Decision probability tensor $\mathbf{D}(i, j, k)$ with dimension $n(i, j, k)$

The dimensionality of these local tensors varies across the grid according to the neuron density, creating a discretized approximation of the continuous tensor fields described in the theoretical framework.

6.3 Discretized Dynamics

The continuous field equations are discretized in both space and time, yielding update rules such as:

$$\mathbf{W}(i, j, k, t + \Delta t) = \mathbf{W}(i, j, k, t) + \Delta t \cdot \frac{1}{\ln 2} \mathbf{Z}(i, j, k, t) \odot \mathbf{D}'(i, j, k, t) \otimes \mathbf{X}(i, j, k, t) \quad (13)$$

Similarly, the local entropy density is computed as:

$$h(i, j, k) = -\frac{1}{\ln 2} \mathbf{Z}(i, j, k) \cdot \Delta \mathbf{D}(i, j, k) \quad (14)$$

6.4 Numerical Computation of Spatial Entropy Gradient

The spatial entropy gradient is fundamental to the SKA Neural Fields framework, directly determining how information flows between neural units through probabilistic connectivity. Computing this gradient presents a unique challenge due to the varying neuron densities across the field, which create discontinuities in the dimensions of the knowledge and decision tensors. To address this, we employ the finite element method (FEM)—a powerful, mesh-based technique widely used in computational physics and engineering—which naturally accommodates spatial variations in computational properties. This approach discretizes the continuous neural field

into tetrahedral elements, allowing us to compute accurate entropy gradients even across regions with significantly different neuron counts. Unlike analytical formulations that struggle with dimensional transitions, the FEM provides a mathematically rigorous foundation for capturing how entropy varies throughout the three-dimensional neural medium.

The above methodology extends naturally to higher-dimensional neural fields (e.g., 4D or 5D) by replacing tetrahedral elements with their D -simplex counterparts (4-simplex in 4D, 5-simplex in 5D), computing gradients as D -dimensional tensors, and using hypervolume-weighted averaging for nodal recovery. All assembly steps, shape functions, and Jacobian transformations generalize directly to arbitrary dimension D .

6.4.1 Scalar Field at Mesh Nodes

We denote by h_h the finite-element approximation of the continuous entropy field h , where the subscript h indicates dependence on the mesh discretization (mesh size parameter).

Let each node n carry

$$\mathbf{z}_n \in \mathbb{R}^d, \quad \mathbf{D}_n(t) \in \mathbb{R}^d,$$

with d the dimension of the knowledge/decision tensors. Define the one-step change

$$\Delta \mathbf{D}_n = \mathbf{D}_n(t + \Delta t) - \mathbf{D}_n(t).$$

We compute the scalar entropy value at each global node:

$$H_n = -\frac{1}{\ln 2} \mathbf{z}_n \cdot \Delta \mathbf{D}_n. \quad (15)$$

Then interpolate onto the continuous field via the FEM shape functions $\{\phi_n(\mathbf{r})\}_{n=1}^N$:

$$h_h(\mathbf{r}) = \sum_{n=1}^N H_n \phi_n(\mathbf{r}), \quad (16)$$

where N is the total number of mesh nodes.

6.4.2 Element-wise Gradient Computation

For linear tetrahedral elements, $\nabla \phi_a$ is constant within each element K [7]. Let $I[a]$ map the a th local node of K to its global index. At element K ,

$$\nabla_{\xi} \phi_a \longrightarrow \nabla \phi_a = J_K^{-T} \nabla_{\xi} \phi_a,$$

with J_K the Jacobian of the element mapping and $|K| = \frac{1}{6} |\det J_K|$. The element-wise entropy gradient is then

$$\nabla h_K = \sum_{a=1}^4 H_{I[a]} \nabla \phi_a, \quad \text{constant on } K. \quad (17)$$

6.4.3 Nodal Gradient Recovery

Having ∇h_K for each element, we recover a nodal gradient $\mathbf{g}_n \in \mathbb{R}^3$ by volume-weighted averaging:

$$\mathbf{g}_n = \frac{1}{\sum_{K \ni n} |K|} \sum_{K \ni n} |K| \nabla h_K, \quad (18)$$

where the sum is over all tetrahedra K containing node n .

6.4.4 Initialization of Entropy Gradients in Neural Fields

The learning process begins with a single activated SKA unit at node n_0 . We set

$$H_{n_0} = -\frac{1}{\ln 2} \mathbf{z}_{n_0} \cdot \Delta \mathbf{D}_{n_0},$$

and for all other nodes $n \neq n_0$, $H_n = 0$. The scalar field $h_h(\mathbf{r})$ is then assembled via FEM shape functions.

Applying the element-wise and nodal-recovery steps yields

$$\mathbf{g}_n = \frac{1}{\sum_{K \ni n} |K|} \sum_{K \ni n} |K| \nabla h_K,$$

where only tetrahedra incident on n_0 contribute nonzero ∇h_K . As a result, exactly the immediate mesh neighbors of n_0 have nonzero $\mathbf{g}_n$; all others are zero.

These initial gradient components determine the first connectivity pattern, directing information flow from the activated unit to its neighbors.

As each newly activated unit calculates its own entropy in subsequent learning steps, a more complex entropy landscape develops, with non-uniform gradients across the grid. This creates a self-organizing connectivity pattern that evolves dynamically through the learning process, without requiring predetermined connections.

This initialization approach elegantly bootstraps the entire learning process from a single point, allowing the network connectivity to emerge organically through entropy reduction propagation.

7 Riemannian SKA Neural Fields

7.1 Riemannian Manifolds as Natural Learning Spaces

Riemannian geometry provides a mathematical framework that elegantly addresses the limitations of discrete-layered models—such as their inability to represent spatial continuity, heterogeneous neuron densities, and functional anisotropy—by reframing the learning space as a curved manifold where the metric—the notion of distance itself—is determined by information-theoretic properties rather than an imposed Euclidean structure.

A fundamental property of the SKA framework establishes that uniform density regions ($\nabla \rho = 0$) naturally lead to uniform entropy ($\nabla h = 0$). This relationship emerges directly from SKA’s entropy-driven learning dynamics and provides the theoretical foundation for when Riemannian geometry becomes necessary versus when simpler Euclidean approaches suffice.

In a Riemannian framework, the learning space becomes a manifold M equipped with a metric tensor g_{ij} that varies from point to point based on local information properties:

$$g_{ij}(\mathbf{r}) = \alpha \cdot (\nabla h)_i (\nabla h)_j + \beta \cdot (\nabla \rho)_i (\nabla \rho)_j + \gamma \cdot \delta_{ij} \quad (19)$$

Where:

- $(\nabla h)_i$ and $(\nabla h)_j$ are components of the entropy gradient
- $(\nabla \rho)_i$ and $(\nabla \rho)_j$ are components of the density gradient
- α , β , and γ are weighting parameters
- δ_{ij} is the Kronecker delta, ensuring positive-definiteness of the metric.

Justification of the Metric Structure. The metric structure in Equation (19) follows directly from three principles:

1. **Scalar invariance.** Both $h(\mathbf{r})$ and $\rho(\mathbf{r})$ are scalars; therefore their gradients ∇h and $\nabla \rho$ are the lowest-order covariant objects that encode spatial variation. Any Riemannian metric constructed from these fields must depend on tensors of rank 2. The outer products $(\nabla h)_i(\nabla h)_j$ and $(\nabla \rho)_i(\nabla \rho)_j$ are the unique symmetric rank-2 tensors formed from first derivatives.
2. **Minimality of order.** Higher-order derivatives (e.g., Hessians $\partial_i \partial_j h$) would introduce curvature-like information into the metric itself, conflating geometry with its derivatives. SKA requires that curvature emerges from the metric, not inside it. Therefore the metric must depend only on first-order quantities.
3. **Linearity and independence.** The metric must remain positive-definite and must separate entropic variation from density variation. A linear combination with coefficients α, β, γ is the minimal structure that:
 - preserves symmetry
 - preserves positive-definiteness
 - treats h and ρ as independent sources of geometric anisotropy
 - reduces to Euclidean geometry when $\nabla h = \nabla \rho = 0$

Under these constraints, the above expression is the unique lowest-order Riemannian metric compatible with entropy-driven SKA learning. This formulation creates a space where “distance” is defined not by physical separation but by information difference: points with similar entropy and density properties become informationally “closer” regardless of their Euclidean positions.

In this Riemannian formulation, learning paths follow geodesics—locally distance-minimizing curves—according to the geodesic equation:

$$\frac{d^2 x^i}{dt^2} + \Gamma_{jk}^i \frac{dx^j}{dt} \frac{dx^k}{dt} = 0 \quad (20)$$

where:

- x^i are coordinates in the information manifold,
- Γ_{jk}^i are Christoffel symbols derived from the metric tensor $g_{ij}(\mathbf{r})$,
- t parametrizes the learning progression

7.2 Geodesic Learning Paths

In Riemannian geometry, the shortest path between two points is a geodesic, which generalizes the concept of straight lines to curved spaces. For SKA, geodesics represent optimal learning paths that minimize information distance.

The geodesic distance between points a and b is given by:

$$d(a, b) = \min_{\gamma} \int_a^b \sqrt{g_{ij} \frac{dx^i}{dt} \frac{dx^j}{dt}} dt \quad (21)$$

Where the minimum is taken over all paths γ connecting a and b . This represents the information distance in our learning space.

7.3 Geodesic Paths for Continuous Knowledge Propagation

The Riemannian metric

$$g_{ij}(\mathbf{r}) = \alpha \cdot (\nabla h)_i (\nabla h)_j + \beta \cdot (\nabla \rho)_i (\nabla \rho)_j + \gamma \cdot \delta_{ij}$$

governs geodesic paths that enable continuous knowledge propagation by minimizing variations in the gradients of local entropy (∇h) and neuron density ($\nabla \rho$). These geodesics, defined as paths that minimize the information distance (Equation 20), naturally select trajectories along which both ∇h , computed via finite element methods (Section 6.4), and $\nabla \rho$, generated through Simplex noise (Section 3.2), evolve smoothly. This minimal-gradient property avoids abrupt transitions in the information manifold, ensuring that knowledge propagates coherently across regions with differing entropy and computational capacity, in line with observations from biological neural systems [3].

The smoothness of these geodesics reveals the fundamental information-theoretic principles underlying biological neural organization. The weight update rule emerging from the entropic least action principle in our prior SKA framework [1, 2]:

$$\frac{\partial \mathbf{W}(\mathbf{r}, t)}{\partial t} = -\nabla_{\mathbf{z}} H(\mathbf{r}, t) \otimes \mathbf{X}(\mathbf{r}, t)$$

where the entropy gradient with respect to knowledge is given by:

$$\nabla_{\mathbf{z}} H = -\frac{1}{\ln 2} \mathbf{z}(\mathbf{r}, t) \odot \mathbf{D}'(\mathbf{r}, t)$$

with $\mathbf{D}'(\mathbf{r}, t) = \mathbf{D}(\mathbf{r}, t) \odot (1 - \mathbf{D}(\mathbf{r}, t))$ being the sigmoid derivative. Note that $\nabla_{\mathbf{z}} H$ has the same dimensionality as the decision probability tensor $\mathbf{D}$, representing the rate of entropy change per unit knowledge in each representational dimension. This structure exhibits an outer product $\otimes$ that mathematically resembles correlation-based learning mechanisms. However, this similarity is superficial—the driving force is fundamentally different. Where classical correlation-based learning strengthens connections between co-active elements, SKA’s outer product represents the alignment of entropy gradients ($\nabla_{\mathbf{z}} H$) with input information ($\mathbf{X}$). The gradient $\nabla_{\mathbf{z}} H$ is not neural activity but rather the information-theoretic direction that guides optimal knowledge organization through entropy minimization.

This distinction reveals that what appears as Hebbian-like connectivity in biological systems may actually reflect deeper information-theoretic optimization. Neural systems that evolved to minimize entropy naturally develop connection patterns that correlate inputs with entropy-reducing responses, creating the appearance of activity-based correlation while actually following information-geometric principles. The ubiquitous biological observation of correlated activity strengthening connections [19] emerges as a surface manifestation of entropy minimization in information systems, not as an independent learning rule.

The temporal consistency of gradient-driven paths naturally produces spike-timing-dependent-like effects [20]. In SKA Riemannian Neural Fields, the sequential activation of units along geodesic paths creates temporal correlations that strengthen information-optimal connections through the entropy minimization process. This temporal ordering emerges from the geodesic structure itself, where the principle of least information distance naturally creates timing relationships while maintaining forward-only learning dynamics. This process is implemented computationally through discrete exterior calculus (Section 7.6), where edge weights

$$w_{ij} = \sqrt{g_{uv} \Delta x_{ij}^u \Delta x_{ij}^v}$$

encode local metric variations, enabling efficient geodesic computation using `pydec` and `FEniCS`. By prioritizing information-optimal paths that minimize entropy and density gradients, the metric tensor defines a biologically plausible and computationally scalable mechanism for self-organizing connectivity in continuous neural fields.

7.4 Boundary Condition for the Learning Path

The learning path in SKA Neural Fields is governed not only by its initialization but also by critical boundary conditions that shape its evolution. While the initialization process establishes the initial entropy gradient landscape radiating from a seed point, boundary conditions determine how knowledge accumulates and flows throughout the field as learning progresses.

The two essential boundary conditions for the learning path are:

1. **Starting Point Condition:** The learning path originates from regions of highest neuron density within the field. This boundary condition leverages the computational capacity of high-density regions to initiate structured knowledge accumulation. By starting at density maxima, the system naturally exploits areas with greater representational capability to establish initial knowledge structures, which then propagate to regions with lower capacity.
2. **Endpoint Density Condition:** Rather than explicitly defining fixed endpoint locations, we specify an endpoint density characteristic. This enables the system to autonomously discover which points in the neural field satisfy this condition. The learning path self-organizes between the high-density starting points and these emergent endpoints, creating optimal pathways for tasks such as classification.

Importantly, under these boundary conditions, the learning trajectory realized by the SKA Neural Field is formally a *geodesic* in the information manifold defined by the Riemannian metric tensor $g_{ij}(\mathbf{r})$. This means that knowledge propagates along locally entropy-minimizing (information-optimal) paths connecting the high-density origin regions to emergent endpoint regions, with the path itself adapting to the evolving entropy and density landscape.

These boundary conditions enable the spatial entropy gradient to continuously update as knowledge accumulates, guiding information flow along geodesics that locally and globally minimize entropy. This self-organizing approach maintains the forward-only learning principle while allowing for flexible, task-specific pathway formation.

When applied to classification tasks, this framework enables the system to not only reduce entropy locally but to do so in a way that naturally develops geodesic connections between high-density processing regions and appropriate classification output regions. The system’s ability to discover optimal geodesic learning paths between boundary-defined regions, rather than following predetermined routes, represents a fundamental advantage over traditional neural architectures with fixed connectivity patterns.

7.5 The Duality of Local and Spatial Learning Paths

The SKA Neural Fields framework reveals a profound duality in how knowledge accumulates, governed by two fundamental principles: locally, knowledge evolves temporally following the **first law of entropic least action**; globally, knowledge propagates spatially following the **second law of spatial least action** across the entropy and density landscapes. These complementary learning principles operate in parallel, manifesting over distinct axes of time and space.

7.5.1 Local Temporal Learning Path

At each fixed spatial position $\mathbf{r}$ in the neural field, knowledge evolves temporally according to the local entropic least action principle (the first law of SKA learning). The action integral for local learning is given by:

$$H(\mathbf{r}) = \frac{1}{\ln 2} \int L(\mathbf{r}, t) dt \quad (22)$$

where $L(\mathbf{r}, t)$ is the local Lagrangian (e.g., the local entropy rate). The resulting temporal trajectory describes how knowledge is progressively accumulated at each location, with learning following the optimal entropy-minimizing path in time.

7.5.2 Spatial Propagation Learning Path

Simultaneously, guided by the second law of SKA learning, the accumulation of knowledge propagates spatially across the field along geodesic paths in the information manifold. These paths represent routes of optimal information efficiency. This spatial propagation is governed by the Riemannian metric $g_{ij}(\mathbf{r})$ and follows the geodesic equation:

$$d(a, b) = \min_{\gamma} \int_a^b \sqrt{g_{ij} \frac{dx^i}{dt} \frac{dx^j}{dt}} dt \quad (23)$$

Importantly, the integrand

$$L = \sqrt{g_{ij} \frac{dx^i}{dt} \frac{dx^j}{dt}}$$

serves as the **Lagrangian** for the spatial learning path. The geodesic trajectory γ is thus obtained by minimizing the action functional

$$S[\gamma] = \int_a^b L dt.$$

Minimizing this action using the Euler–Lagrange equations yields the geodesic equation:

$$\frac{d^2 x^i}{dt^2} + \Gamma_{jk}^i \frac{dx^j}{dt} \frac{dx^k}{dt} = 0,$$

where Γ_{jk}^i are the Christoffel symbols derived from the metric g_{ij} . This makes clear that the spatial propagation of knowledge in Riemannian SKA neural fields is governed by a *variational principle*, fully analogous to the temporal least action principle at each point. Thus, the learning path is not merely an optimal route by construction, but is dynamically discovered through the minimization of the action functional defined by the information geometry of the neural field.

The spatial learning path γ connects regions (for example, from a high-density starting point to an endpoint defined by boundary conditions) and represents an optimal route that minimizes information distance—i.e., a **geodesic** in the information manifold.

7.6 Learning Path Realization in Riemannian SKA Neural Fields

The process by which the learning path is established and executed in Riemannian SKA Neural Fields proceeds as follows:

1. **Initialization of Entropy Fields:** The learning process is initiated by constructing the entropy field across the neural domain. As detailed in the subsection 6.4.4, a seed node (typically in a region of maximal neuron density) is activated. The initial local entropy values are computed, creating the first entropy gradient landscape which serves as the informational scaffold for subsequent knowledge propagation.
2. **Specification of Boundary Conditions:** Boundary conditions are established by defining the start and end regions of the learning path. As detailed in subsection 7.4, the origin is generally set at a density maximum (starting point), while the endpoint is determined by a neuron density threshold or by task-driven criteria (e.g., a classification region). *In cases where the endpoint density criterion admits multiple possible locations within the field, the set of all eligible endpoints is considered.*

3. **Computation of the Geodesic Path:** Given the entropy field and boundary conditions, the geodesic—the information-optimal path—is computed on the Riemannian manifold. At this stage, the entropy field is typically sparse, having been initialized from a single activated seed node. Since the final endpoint is not known a priori, a set of candidate endpoints (usually located in regions of comparable neuron density) is defined. A geodesic is computed from the seed to each candidate, and the path with the minimal information distance is selected. This procedure identifies the most efficient endpoint and defines the initial trajectory for learning. The selected path minimizes the information distance defined by the local metric $g_{ij}(\mathbf{r})$ and aligns naturally with both entropy and density gradients.
4. **Forward Pass ($K = 1$):** A forward pass is performed along the computed geodesic path. At each node traversed by the geodesic, knowledge and decision probability tensors are updated according to the entropic least action principle. This forward pass corresponds to the first step ($K = 1$) in the learning path and embodies the SKA’s forward-only update mechanism.
5. **Update of Entropy Fields:** Following the forward pass, the entropy field across the domain is recomputed to incorporate the effects of the newly acquired knowledge. The entropy gradient landscape is thus dynamically updated, influencing the formation of subsequent geodesic paths for continued learning.

This iterative procedure allows the SKA Neural Field to adaptively self-organize, accumulating knowledge in a manner that is both locally entropy-minimizing and globally information-optimal. The interplay between local updates and global geodesic propagation is fundamental to the SKA paradigm’s ability to realize scalable and efficient learning in continuous neural fields.

Convergence is achieved when, after a finite number of forward passes (for some optimal K^*), the geodesic path stabilizes and remains invariant under further iterations. At this stage, the local entropy values $h_i^{K^*}$ along the geodesic also converge, indicating that knowledge accumulation and entropy reduction have reached a steady state along the optimal information pathway. Thus, completion of the learning path is formally marked by the invariance of the geodesic and the convergence of local entropy values along it.

In essence, the Riemannian SKA Neural Fields framework gives us the optimal architecture for the local learning path, allowing the system to discover the most efficient connectivity pattern through entropy and density gradient guidance rather than relying on predetermined architectural choices. This represents a fundamental advance over traditional approaches where architecture design precedes learning.

Figure 2 illustrates how learning paths evolve over successive forward passes as the entropy landscape is updated, demonstrating the dynamic nature of geodesic formation in the Riemannian SKA Neural Field. The temporal separation between successive geodesics is governed by the characteristic time step Δt , which represents the fundamental weight update interval that allows the entropy field to reorganize and establish new optimal pathways for knowledge propagation. As the entropy evolves and converges to equilibrium, the difference between successive geodesics vanishes, indicating that the system has completed its architecture discovery process and identified the optimal neural connectivity pattern for information flow—marking the convergence of both the temporal and spatial learning paths established in the duality framework above (Section 7.5).

Geodesic Learning Paths

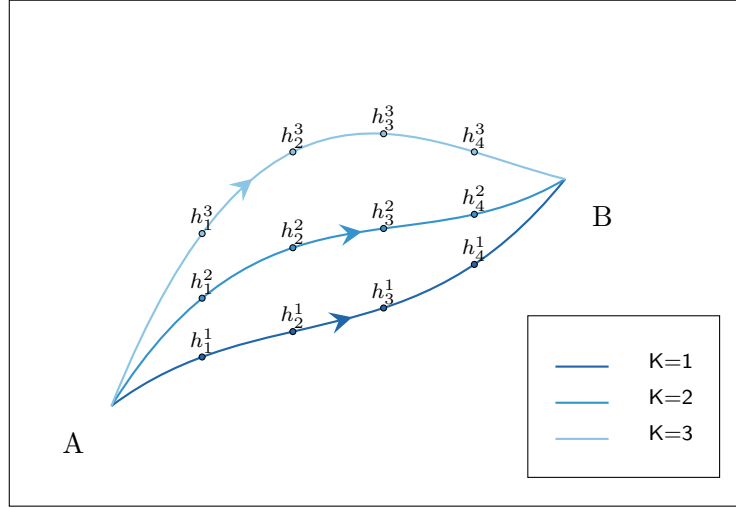


Figure 2: Three elevated geodesic learning paths connecting points A and B for different forward passes ($K = 1, 2, 3$). Each path shows local entropy values h_i^K at intermediate points. As K increases, the paths rise and curve more to optimize information flow according to the evolving entropy landscape. Arrows indicate mid-path knowledge direction.

7.7 Computational Implementation via Discrete Exterior Calculus

Implementing continuous Riemannian geometry for the SKA Neural Fields framework presents significant computational challenges due to the complexity of high-dimensional manifolds and varying neuron density. Discrete exterior calculus (DEC), implemented via the Python library `pydec`, provides a computationally tractable framework that preserves differential geometric properties. This subsection outlines the DEC-based pipeline, integrating `pydec` for discrete geodesic and differential form computations, `FEniCS` for finite element method (FEM) calculations, `TorchManifolds` for differentiable Riemannian metric and geodesic operations, `noise` for neuron density generation, and `PyTorch` for numerical tensor operations.

The key elements of the DEC implementation include:

1. **Simplicial Complex:** A tetrahedral mesh (or D-simplex mesh for higher dimensions) adapts to the information landscape, constructed using `pydec`'s simplicial complex tools, with vertices representing computational units.
2. **Discrete Differential Forms:** Gradients (e.g., entropy gradient ∇h) and tensor fields (e.g., knowledge tensor $\mathbf{Z}$, decision probability tensor $\mathbf{D}$) are represented as discrete forms, leveraging `pydec`'s cochain framework.
3. **Hodge Star Operator:** `pydec` implements the Hodge star operator to compute gradients and divergences, enabling accurate entropy gradient calculations (Section 6.4.3) without continuous representations.
4. **Discrete Geodesics:** Shortest paths in the information manifold, representing optimal learning trajectories, are computed using `pydec`'s weighted graph utilities.

To compute a geodesic path between units a and b :

1. **Initialization:** Triangulate the neural field into a tetrahedral mesh using `pydec`, with vertices corresponding to computational units. Neuron density $\rho(\mathbf{r})$ is generated using `noise`'s Simplex noise (Section 3.2).

2. **Metric Assignment:** For each edge (i, j) , assign a weight:

$$w_{ij} = \sqrt{g_{uv} \Delta x_{ij}^u \Delta x_{ij}^v} \quad (24)$$

where Δx_{ij}^u is the coordinate difference in dimension u , and g_{uv} is the metric tensor (Equation 19) evaluated at the edge midpoint using **TorchManifolds**. The metric incorporates entropy gradients (∇h) from **FEniCS** (Section 6.4) and density gradients $(\nabla \rho)$ from the noise-generated field.

3. **Path Computation:** Apply Dijkstra’s algorithm in **pydec** to find the shortest path from a to b , representing the information-optimal learning path (Section 7.2).
4. **Gradient Flow:** Update knowledge and decision probability tensors $(\mathbf{Z}, \mathbf{D})$ along the geodesic using the entropic least action principle (Equations 8, 13), with tensor operations (e.g., $\odot$, $\otimes$) performed using **PyTorch**. Recompute the entropy field $h_h(\mathbf{r})$ using **FEniCS**’s FEM shape functions (Section 6.4.1).

The pipeline integrates:

- **FEniCS:** Interpolates scalar entropy values H_n , computed by **PyTorch**, into the continuous entropy field $h_h(\mathbf{r})$ and calculates its gradient ∇h_K on tetrahedral elements, providing gradients for **pydec**’s discrete operations.
- **TorchManifolds:** computes the Riemannian metric tensor and Christoffel symbols automatically using **PyTorch** autograd $g_{ij}(\mathbf{r})$ (Equation 19), supporting continuous geodesic validation.
- **pydec:** Computes discrete geodesics and differential operations, using **FEniCS**-computed gradients to bridge continuous and discretized domains.

Computational dependencies include:

- **pydec:** Mandatory for DEC operations (simplicial mesh, discrete forms, geodesics).
- **FEniCS:** Essential for FEM-based entropy gradient calculations.
- **TorchManifolds:** Critical for Riemannian metric tensor manipulation.
- **noise:** Required for Simplex noise-based neuron density generation.
- **PyTorch:** Necessary for tensor operations (e.g., $\mathbf{Z} \odot \mathbf{D}'$, $\mathbf{Z} \cdot \Delta \mathbf{D}$).

Using only **FEniCS** and **TorchManifolds** is infeasible, as **pydec** is required for discrete geodesic computations on simplicial meshes, which neither **FEniCS** (FEM-focused) nor **TorchManifolds** (continuous geometry) can provide. For higher dimensions (e.g., 4D, 5D), **pydec** extends to D-simplex complexes, **FEniCS** adapts FEM, and **TorchManifolds** generalizes the metric tensor. We draw upon Chattopadhyay [4] for **TorchManifolds** and Hughes [7] for **FEniCS**.

7.8 Dimensional Integration and Functional Modulation

A key advantage of the Riemannian approach is its ability to unify physical space with additional informational or functional properties within a single geometric framework. Traditional models often treat these properties—such as attention, abstraction, or contextual influence—as separate computational layers or modules. In contrast, the Riemannian SKA Neural Field embeds them directly into the metric structure, allowing these properties to modulate geometry without introducing new spatial coordinates.

Let $f_k(\mathbf{r})$ denote a set of scalar functional fields that capture local informational states or modulatory factors influencing learning. These fields modify the metric coefficients according to:

$$g_{ij}(\mathbf{r}) = \alpha (\nabla h)_i (\nabla h)_j + \beta (\nabla \rho)_i (\nabla \rho)_j + \sum_{k=1}^n \gamma_k (\nabla f_k)_i (\nabla f_k)_j + \gamma_0 \delta_{ij}. \quad (25)$$

Here, $(\nabla f_k)_i$ represents the i -th component of the gradient of the k -th functional field, influencing the local curvature and connectivity of the manifold. These fields do not expand the coordinate dimension of the neural manifold but instead act as geometric modulators that reshape the information metric.

This formulation mitigates the exponential scaling that would occur if each new property required an explicit higher-dimensional representation. The Riemannian framework thus encodes complex functionality through metric modulation on a fixed- D manifold, with computational complexity that scales linearly with the number of modulatory fields.

Clarification on Dimensionality. In this work, $D \in \{3, 4, 5\}$ denotes the ambient dimension of the manifold on which the neural field is defined. Introducing additional scalar fields f_k does not increase the coordinate dimension; instead, these fields modulate the coefficients of the Riemannian metric $g_{ij}(\mathbf{r})$, thereby shaping the geometry on a fixed- D manifold. Accordingly, our FEM/DEC formulations extend to D -simplices (e.g., tetrahedra in 3D, 4-simplices in 4D) without introducing extra coordinates.

8 Future Work

While this paper establishes the theoretical foundations of Riemannian SKA Neural Fields, extending the discrete SKA framework to continuous, geometrically principled systems, several avenues remain for practical validation and enhancement. Future research will focus on implementing the proposed framework using the outlined computational pipeline, including finite element methods (FEM) and discrete exterior calculus (DEC) via libraries such as `pydec`, `FEniCS`, and `TorchManifolds`.

Additionally, we aim to explore higher-dimensional generalizations ($D = 4, 5$) for modeling complex spatial spaces and to extend the mechanisms of entropy-driven architecture emergence to more heterogeneous and dynamic fields. Because the SKA entropy expression is independent of spatial dimension, these extensions remain mathematically natural. A subsequent paper will detail these implementations, experimental results, and performance metrics, demonstrating scalability for real-time, resource-efficient AI systems.

These directions aim to empirically validate the theoretical claims presented here and demonstrate the practical viability of Riemannian SKA Neural Fields in large-scale learning systems.

9 Conclusion

In this work, we introduced Riemannian SKA Neural Fields, a geometric extension of the Structured Knowledge Accumulation (SKA) framework to continuous neural systems. By modeling the neural medium as a Riemannian manifold with a metric tensor encoding entropy and density gradients, we enable knowledge propagation along geodesics, balancing information structuring with spatial heterogeneity. This approach generalizes the entropic least action principle from discrete layered networks to spatially continuous fields, providing a unified, biologically plausible paradigm for adaptive learning in complex architectures.

Our formulation bridges information theory, differential geometry, and neural computation, addressing limitations in prior SKA works by incorporating anisotropic connectivity and functional organization. Through finite element discretization and discrete geometric computation,

the framework offers scalable tools for higher-dimensional simulations, opening new pathways for efficient AI design. While theoretical in scope, this foundation paves the way for practical applications in neuromorphic computing and self-organizing systems, with empirical validations to follow in future studies. The implementation and evaluation of these principles constitute the next step toward realizing self-organizing, entropy-guided architectures in practice.

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